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SURVEY

IQA Vision Transformed: A Survey of Transformer Architectures in Perceptual Image Quality Assessment

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ABSTRACT In an era dominated by visual content, perceptual image quality assessment (IQA) is crucial for enhancing user experiences and driving technological advancements across various domains. This survey paper reviews the integration of Vision Transformers (ViTs) into both no-reference (NR) and full-reference (FR) IQA methods, highlighting their promise as alternatives to traditional techniques. ViTs leverage attention mechanisms to focus selectively on relevant image patches, showing promise in aligning more closely with human perceptual errors. We identify key limitations of conventional IQA methods and track the evolution from early learning-based approaches to contemporary deep learning models, with a specific focus on ViTs. We discuss the performance of Transformer-based models in capturing image distortions and their strong correlation with subjective IQA metrics. We also discuss potential advancements, including the development of hybrid architectures combining diverse deep learning approaches, adaptive IQA mechanisms through meta-learning, and scalable solutions inspired by emerging computational paradigms. These advancements promise to enhance perceptual quality assessment, with substantial implications for industries such as medical imaging, multimedia applications, and beyond. This study aims to set the groundwork for future research in transformer-based methodologies, offering new insights into the transformative impact of these models on IQA.

INDEX TERMS Perceptual image quality assessment (IQA), vision transformers (ViTs), transformer architectures, deep learning for IQA, multimedia applications, cross-domain evaluation, attention mechanisms, hybrid architectures, quantum-inspired computing, meta-learning for IQA, capsule networks.

I. INTRODUCTION

In an era influenced by visual content, the assessment of perceptual IQA has become pivotal task across diverse domains, influencing user experiences, decision-making processes, and technological advancements. The significance of IQA is particularly critical in real-world applications such as image restoration, compression, and rendering. Within visual communication systems, quality assessment metrics serve as both testing standards and optimization objectives. These metrics are categorized into subjective and objective methods. While subjective assessment is widely regarded as the gold standard by utilizing human visual system (HVS) for IQA, but using human observers for IQA is hindered due to being time-consuming, expensive, and impractical for direct integration into real-time systems. In contrast, objective quality evaluation systems are often developed or trained using subjective assessment data that offer an automated

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means of predicting visual quality. These objective methods prove invaluable for swift evaluation and enhancement of images.

Over the past several decades, the Mean Squared Error (MSE) stood as the conventional full-reference method for gauging signal fidelity and quality [1], [2], [3]. Despite its enduring role in shaping signal and image processing algorithms, the MSE's limitations are evident in its weak correlation with human perception [1], [4].

Convolutional Neural Networks (CNNs) have made substantial contributions to the field of computer vision, enabling significant advancements across various applications [5], [6], [7], [8]. The core of CNNs is rooted in the use of the convolution operator, which operates locally and ensures translational equivariance. While these characteristics contribute to the effectiveness and versatility of imaging solutions, the local receptive field in convolutional operations imposes limitations on capturing pixel relationships over long distances. Additionally, the stationary weights of convolutional filters remain unchanged for the specific input image content during inference, limiting adaptability. To address the aforementioned challenges, substantial research endeavors have been dedicated to incorporating attention mechanisms [9]. These mechanisms can be viewed as dynamic weight adjustment processes based on input features, seamlessly integrated into the foundational structures of CNNs, aiming to enhance the non-local modeling capability of these structures [10], [11], [12].

The adoption of attention-based 'Transformer' models represents a significant advancement in overcoming these limitations. Transformers, initially introduced for natural language processing, have shown remarkable performance in understanding complex relationships within visual data [13]. Their ability to encode long-range dependencies and capture global features has positioned them as a promising alternative to traditional CNN-based methods [10], [14], [15]. Recent studies have highlighted that Vision Transformers (ViTs) offer a closer alignment with human perceptual errors compared to CNNs, making them particularly appealing for IQA [16], [17], [18], [19].

For instance, the work by Tang et al. demonstrated that integrating transformers into IQA models improved performance across diverse datasets and distortion types, surpassing traditional methods in capturing perceptual quality [20]. Similarly, Chubarau et al. explored the use of ViTs in multimedia applications, showcasing their versatility and effectiveness in handling various distortions [21].

In recent times, significant advancements have been made in the field of IQA through various approaches. Comprehensive surveys such as Lepcha et al.'s review of image super-resolution techniques [22], Wang et al.'s exploration of no-reference IQA based on deep learning [23], Yang et al.'s survey on deep learning-based IQA [24], and Alaei et al.'s review on document image quality assessment [25] provide valuable insights into the evolving landscape of IQA methodologies and their applications.

Despite these extensive reviews, there remains a notable gap in the literature regarding transformer-based IQA techniques. While the aforementioned surveys cover a broad spectrum of IQA approaches and benchmarks, a dedicated survey focusing specifically on transformer-based IQA methods is absent. This omission underscores the need for further investigation into this emerging area. This article explores the complex interactions that exist between perceptual IQA and transformer topologies. Beyond adding to the ever-changing field of computer vision, its goal is to foresee future breakthroughs that may have far-reaching effects on a variety of businesses. The implications of this investigation go into domains including medical imaging and multimedia applications, highlighting the potential wide-ranging effects of developments in transformer-based methodologies.

Our manuscript presents the first comprehensive survey that explores existing work on transformer-based architectures in perceptual IQA. The response of transformer architectures to diverse datasets and degradation levels demonstrates their adaptability and effectiveness in addressing various image quality challenges. By unraveling these complexities, our work sparks inspiration for future research, setting the stage for a informative journey into the uncharted territories of IQA with transformers. The main contributions of this manuscript are as follows,

1) Our work stands as the first comprehensive review of transformer architectures in the domain of Perceptual IQA, encompassing both Full-Reference (FR) and No-Reference (NR) methods. This review provides a detailed examination of transformer-based IQA approaches, positioning them within the broader research landscape and highlighting their advantages over traditional methods.
2) We provide an in-depth analysis of the unique aspects of transformer model structures applied to IQA, including their ability to address local inductive biases and capture global image features, offering a deeper understanding of how these models function in assessing image quality.
3) The survey emphasizes the importance of spatial properties in IQA, illustrating how transformers can overcome challenges related to local feature extraction and effectively model complex pixel relationships, thus enhancing image quality predictions.
4) We conduct a comprehensive comparison of existing datasets, categorizing them into synthetic and authentic subsets, and evaluate how transformer models respond to different types of distortions. This analysis sheds light on the adaptability of transformer-based methods to diverse real-world scenarios.
5) Our manuscript does not only underscore the successes of transformer-based models in Perceptual IQA but also addresses existing challenges. We propose strategies like attention refinement, dataset augmentation, multi-modal data fusion, and adversarial training to

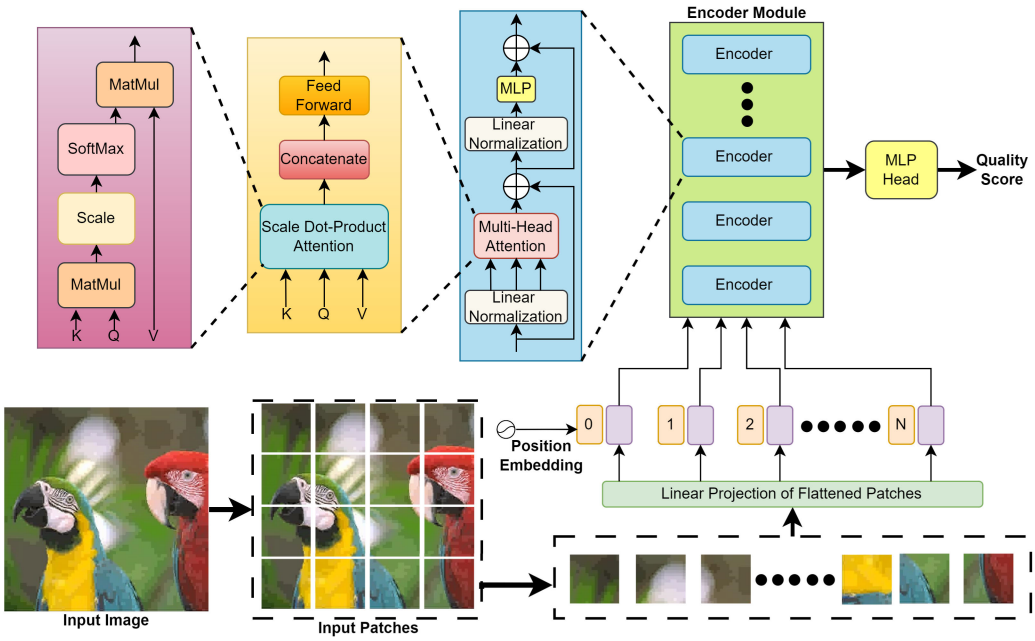

FIGURE 1. Delving into the intricacies of standard vision transformer for perceptual IQA. In this detailed schematic, each facet of the transformer's structure is meticulously elucidated. Witness the symphony of self-attention mechanisms that highlight the power of vision transformers in deciphering and evaluating image quality nuances while Queries (Q), Keys (K), and Values (V) showcase their collaborative prowess in orchestrating self-attention mechanism.

improve these models' performance and generalization capabilities.

6) This survey synthesizes existing findings on the performance of transformer architectures in perceptual IQA, providing a consolidated view that guides researchers in understanding the progress and potential gaps in this evolving field.
7) We outline promising directions for future research in transformer-based IQA, offering insights into potential innovations and setting the stage for new developments that can advance perceptual IQA further.

II. BACKGROUND

This section provides a comprehensive study of IQA, navigating through traditional approaches that have long been cornerstones of image quality evaluation. Following this, we dig into the transformative influence of neural network methods on IQA, unraveling their capacity to discern intricate image features. While this section focuses on conventional and neural network-based IQA methodologies, the subsequent sections will venture into the realm of ViTs and their innovative working mechanisms. By traversing these established and modern approaches, we aim to lay a robust groundwork for understanding IQA evolution.

A. ATTENTION AND TRANSFORMERS

The original application of the Transformer architecture marked a significant advancement in machine translation by introducing a novel attention-driven building block [9]. The fundamental structure of the Transformer comprises an encoder and a decoder, each consisting of a stacked consecutive identical blocks. In contrast to convolutional models, the Transformer exclusively depends on the self-attention mechanism, which is essential for establishing relationships between various points in a sequence in order to calculate the overall representation of the sequence [26].

The distinctive self-attention technique remains a defining feature of Transformer models. Consequently, it is essential to dig into the core function of the attention mechanism and multi-head attention mechanism to comprehend its role and significance in these models.

1) SELF-ATTENTION MECHANISM

The self-attention mechanism is a pivotal component in the transformer architecture, allowing the model to weigh different positions in the input sequence differently, thereby capturing long-range dependencies effectively. In a standard transformer, the self-attention mechanism is employed within each attention layer. Given an input sequence X, the self-attention mechanism calculates attention scores for each position, determining the importance of other positions relative to the current one. This allows the model to concentrate on relevant context, which makes it easier to capture dependencies over different distances.

The self-attention operation involves three linear transformations to obtain query (Q), key (K), and value (V) matrices from the input sequence X. These matrices are used to compute the attention scores, from which a weighted

total yields the self-attention output. In Figure 1, we present a detailed overview of the standard vision transformer architecture, illustrating the nuanced roles of Q, K, and V in orchestrating self-attention mechanisms for enhanced IQA.

The following is a mathematical definition of the self-attention operation in a transformer:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \tag{1}$$

$$\text{scores} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right), \quad \text{weighted_sum} = \text{scores}(V), \tag{2}$$

where d_k denotes the dimensionality of the key vectors and W_Q, W_K, and W_V are learnable linear transformation matrices.

The self-attention mechanism's ability to weigh different elements of a sequence dynamically, based on their relevance to each other, has proven highly effective in capturing long-range dependencies. As a result, transformers have proven successful in a variety of applications, such as computer vision and natural language processing, where precise processing and representation depend on an awareness of the context and relationships between parts.

2) MULTI-HEAD ATTENTION MECHANISM

The multi-head attention technique was developed by Vaswani et al. [9] in order to capture subtle interactions among token entities from various angles. The multi-head attention mechanism, which is at the heart of the transformer architecture, enhances the model's capacity to learn complex relationships by using different attention heads to attend to numerous positions at once.

The multi-head attention mechanism of the traditional transformer is integrated into both the encoder and decoder blocks. The ability of each attention head to function independently allows the model to concentrate on different input stream segments. The outputs of these several attention heads are concatenated and linearly transformed to create the final attention output.

In each attention head, attention scores are computed by measuring the relationship between input vectors Q (queries), K (keys), and V (values) using the formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{3}$$

Here, d_k represents the dimensionality of the key vectors, and the softmax function normalizes the attention scores to determine the weighted sum.

Algorithm 1 describes the algorithm for the multi-head attention mechanism. The transformer can handle both short- and long-range dependencies due to its multi-head attention capabilities, which allows it to identify intricate correlations and patterns in the input sequence. It's success in a variety of computer vision and natural language processing applications can be attributed to its adaptability.

Algorithm 1 Multi-Head Attention Mechanism

Data: Input vectors Q, K, and V; Number of heads H
Result: Multi-head attention output
Initialization:
$d_k \leftarrow$ dimensionality of key vectors;
$W_Q, W_K, W_V \leftarrow$ Learnable linear transformation matrices;
Operation for each head:
for $i \leftarrow 1$ **to** H **do**
  $Q_i \leftarrow QW_{Q_i}$;
  $K_i \leftarrow KW_{K_i}$;
  $V_i \leftarrow VW_{V_i}$;
  $A_i \leftarrow \text{Attention}(Q_i, K_i, V_i, d_k)$;
Concatenation:
$A \leftarrow \text{Concatenate}(A_1, A_2, \ldots, A_H)$;
Linear Transformation:
$Z \leftarrow AW_O$;
Return:
Multi-head attention output Z;

Function `Attention`(Q, K, V, d_k)**:**
  scores $\leftarrow$ Softmax $\left(\frac{QK^T}{\sqrt{d_k}}\right)$;
  weighted_sum $\leftarrow$ scores$V$;
  return weighted_sum;

3) VISION TRANSFORMERS

It has been demonstrated that Transformer models perform well on a variety of natural language processing tasks [27]. Dosovitskiy et al. built on this success by introducing the Vision Transformer (ViT) model in 2020 [10]. Figure 1 depicts the standard vision transformer used for IQA. Remarkably, when trained on huge datasets like JFT-300M, ViT outperformed the current state-of-the-art models, including ResNet-based models like BiT [28].

The ViT method converts images into fixed-sized patches, which are subsequently compressed into vectors. These vectors pass through a layer of trainable linear projection, producing N vectors with D dimensionality, where N is the number of patches. By adding positional embeddings, the outputs at this point, referred to as patch embeddings, maintain positional information. Before the patch embeddings reach the Transformer encoder, a trainable class embedding is also added.

The Transformer encoder, as described in Algorithm 2, comprises multiple encoder blocks, each housing a multi-head self-attention (MSA) block and an MLP block. Before entering these blocks, the activations undergo normalization using LayerNorm (LN). Further enhancing the architecture, skip connections precede the LN, adding a duplicate set of activations to the respective MSA or MLP block outputs. The last phase includes an MLP block that maps the output to class predictions, acting as a classification head.

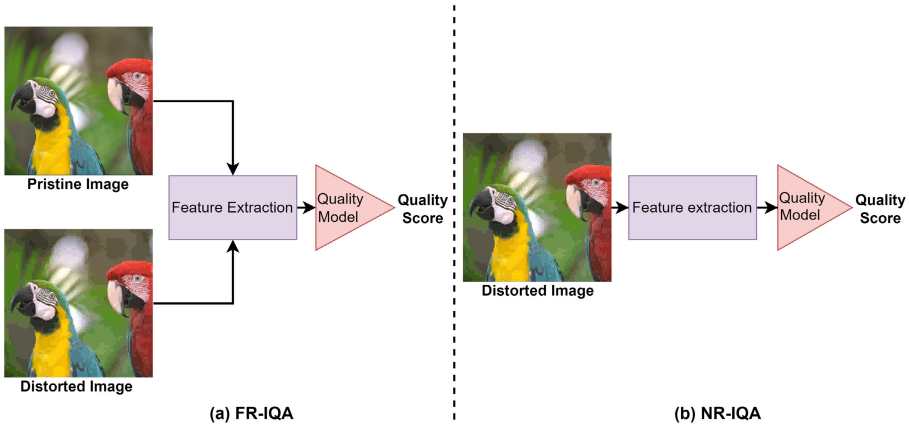

FIGURE 2. Overview of IQA techniques: (a) Illustrating the general process of FR-IQA, and (b) Depicting the general process of NR-IQA.

Algorithm 2 Transformer for Image Quality Assessment

Data: Image
Result: Image Quality Score
Initialization:
Define Vision Transformer (ViT) Model;
Initialize Learnable Parameters;
Image Transformation:
Convert Image to Fixed-Sized Patches;
Flatten Patches into Vectors;
Apply Linear Projection Layer to Obtain N Vectors with Dimensionality D;
Add Positional and Class Embeddings to Patch Vectors;
Transformer Encoder:
for *each Transformer Encoder Block* **do**
  Apply Multi-Head Self-Attention (MSA) Block;
  Apply Layer Normalization (LN) for Activation Normalization;
  Apply Feedforward (MLP) Block;
  Add Skip Connections before LN;
Regression Head:
Apply MLP Block as Regression Head;
Map Output to Image Quality Score;
Training:
Backpropagate and Update Learnable Parameters;
Inference:
Predict Image Quality Score for New Images;
Return:
Image Quality Score;

4) SWIN TRANSFORMER

By using a novel hierarchical approach, Liu et al.'s design, the Swin Transformer, reimagines the use of transformers to image data [29]. Conventional transformers handle photos as flat sequences, which might not be the best way to capture characteristics with hierarchy. This restriction is overcome by the Swin Transformer, which divides the image into non-overlapping patches and processes them through a hierarchical processing pipeline. The transformer blocks at each step work on patches at various resolutions, enabling the model to effectively capture both local and global data. Following are some key characteristics of the Swin Transformer,

- Through the use of fixed-size patch division, the Swin Transformer reduces computing complexity and enables concurrent data processing.
- Transformer blocks are used in many phases to process patches at different resolutions, which facilitates the extraction of characteristics at various scales.
- The Swin Transformer uses shifted windows instead of the conventional sliding window technique, which enhances the model's capacity to record a range of data within each patch.
- The Swin Transformer's scalability is improved by its hierarchical structure, which allows it to process images of different sizes without requiring major adjustments.

The Swin Transformer operates by processing input patches through a series of stages, each containing a unique set of transformer blocks. The input patch at step l is referred as $X^{(l)}$, and the self-attention operation within each transformer block is expressed as:

$$Y^{(l)} = \text{Attention}(X^{(l)}W_Q^{(l)}, X^{(l)}W_K^{(l)}, X^{(l)}W_V^{(l)}), \quad (4)$$

where $W_Q^{(l)}$, $W_K^{(l)}$, and $W_V^{(l)}$ are learnable linear transformation matrices specific to stage l.

The Swin Transformer's ultimate output is generated by combining information from multiple stages:

$$\text{Output} = \text{Concatenate}(Y^{(1)}, Y^{(2)}, \ldots, Y^{(L)}), \quad (5)$$

where L is the total number of stages.

Innovative outcomes in a range of computer vision applications, such as object identification and picture classification, have been shown by the Swin Transformer. It is appropriate for a range of applications due to its flexible design, which

includes efficient hierarchical processing and the capacity to record both local and global information.

B. TRADITIONAL APPROACHES FOR IMAGE QUALITY ASSESSMENT

Approaches in the field of IQA can be broadly classified according on the availability of reference images that are free of distortion. Three categories are identified by this classification i.e., IQA techniques with Full Reference (FR), Reduced Reference (RR), and No Reference (NR) [30]. Every category encompasses a distinct method for assessing the perceptual quality of images, according to the type of reference material that is accessible. We explore the unique features of these groups in the sections that follow, providing insights into their methods and uses in the evaluation of visual fidelity. NR-IQA does not require the pristine version of the images for IQA. Whereas, RR-IQA requires partial information regarding the pristine version of the image under evaluation for IQA. FR-IQA requires the field information regarding the pristine version of the image under evaluation for IQA.

Due to the ability of NR-IQA techniques to evaluate visual integrity without the availability of pristine version of the image, the NR-IQA is usually more practical for real-time applications and is also frequently referred to as blind IQA (BIQA) alternatively. Figure 2 depicts the overview of FR-IQA and NR-IQA techniques. Moreover, while exploring the existing literature, to the best of our knowledge, the literature review revealed a notable absence of studies utilizing transformers in the context of RR-IQA. Therefore, RR-IQA techniques are not discussed in this work, reflecting the current state of the field and signaling a potential avenue for future research endeavors.

The core principle underlying conventional IQA methods revolves around the extraction of features that capture quality-related information from an image. These features are closely associated with image quality and the presence of distortions. Following this, researchers typically design a mathematical function to establish a connection between these features and the perceived image quality score. The term "quality-aware" is used to describe features that exhibit a correlation with either image quality or distortion. In the traditional approach, researchers manually create both the features and the mapping function, drawing on their insights into the reference image, distortion, or the HVS.

Conventional FR-IQA methods, such as mean squared error (MSE) and peak signal-to-noise ratio (PSNR), are recognized for their low computational complexity and clear physical interpretations, making them widely employed in various applications. Despite their practicality, these metrics often exhibit limited correlation with human subjective ratings [31]. In response to this limitation, numerous perceptual FR-IQA metrics have been introduced over the past decades, showcasing promising performance in predicting visual quality [32], [33], [34]. However, in many real-world scenarios, access to a reference image is restricted, prompting the need for the development of NR-IQA methods for such scenarios.

In accordance with their applications, existing NR-IQA techniques can be broadly classified into distortion-specific and general-purpose categories. Distortion-specific techniques are specifically tailored for particular application scenarios, addressing specific distortions such as Gaussian blur [35], JPEG compression [36], JPEG 2000 compression [37], contrast distortion [38], and multiple distortions [39]. These techniques assume prior knowledge of the distortion type in the distorted image, limiting their effectiveness to cases where distortion types are known. Consequently, they may prove inadequate for handling online images with unknown distortion types. In contrast, general-purpose techniques aim to assess the image quality degraded by various types of distortions. Hence, BIQA i.e., IQA without the use of a reference image is a challenging task due to the diversified nature of image content and the multitude of image artifacts.

The majority of existing conventional BIQA techniques adopt an opinion-aware approach, employing a two-stage framework. Initially, perceptual features representing image quality are derived to encapsulate the influence of distortions. The relationship between these extracted features and the overall quality of the image is then established through the use of a regression models. The distorted visuals and the corresponding subjective assessments are used to train these models. The effectiveness of these BIQA methods is based on the extracted quality-aware features. Commonly used features include handcrafted features such as image gradients, natural scene statistics (NSS) characteristics, image filter responses, and entropies [39], [40], [41], [42], [43], [44], [45], [46].

Wavelet and NSS-based BIQA approaches such as BIQI [40] and DIIVINE [41] use a two-stage framework to identify distortion classes and then assess the degree of quality corresponding to each distortion. Using a Bayesian inference methodology, BLIINDS2 [42], a DCT NSS-based method, predicts quality scores by extracting statistical features from the DCT domain. Other approaches, such as those presented by Mittal et al. [43], uses generalized Gaussian distribution (GGD) to elucidate normalized luminance coefficients at the pixel level and asymmetric GGD (AGGD) to explain their paired computations. In [44], for accurately assessing the perceptual quality, the technique relies on extraction of combined distributions of Laplacian of Gaussian (LOG) and image gradient magnitudes. Meanwhile, [45] adopts a unique approach by combining spatial domain NSS features with perceptual characteristics influenced by the free energy theory, giving rise to the NFERM technique. NRSL, as presented in [46], calculates luminance and structural histograms in the spatial domain, demonstrating the favorable roles of these features in IQA. Additionally, [39] employs quality-aware perceptual structure characteristics derived from multiple high-order derivative images. Learning-based

features make up the second feature category, which acquire the quality-aware features through the learning process from raw local image patches [47], [48].

Opinion-free BIQA approaches can be classified into two groups: clean model-based methods, which do not require human subjective ratings for training, and methods that use objective scores generated by FR-IQA methods to substitute human subjective evaluations [49], [50], [51]. The distance between features recovered from distorted images and a pristine image model is computed using clean model-based techniques. For instance, in [49], multivariant Gaussian (MVG) model is constructed using NSS characteristics from natural images in order to describe the regularities found in natural images. Xue et al. [44] incorporate three perceptual characteristics from the gradient, log-Gabor, and color domains that improve performance even more. Additionally, some studies train BIQA algorithms using objective scores produced by FR-IQA techniques in place of human subjective judgments [51].

Opinion-aware and opinion-unaware approaches each have pros and cons of their own. Opinion-aware techniques typically outperform their counterparts due to their thoughtful feature design and sophisticated machine learning approaches. However, these methods demand a similarity in distortion types to be present in training and testing datasets, which restricts their ability to generalize over unseen distortion types. On the other hand, clean model-based opinion-unaware approaches, relying solely on pristine images, show greater potential for generalization across various distortion types. Opinion-unaware approaches that replace subjective scores with FR-IQA techniques face similar difficulties as opinion-aware methods because they also require training on distorted images.

C. DEEP LEARNING-BASED METHODS FOR IMAGE QUALITY ASSESSMENT

In recent years, the field of IQA BIQA has witnessed significant advancements, particularly with the emergence of deep learning approaches. Various models leveraging deep neural networks (DNNs) have been developed to forecast image quality. To overcome the constraints of traditional IQA methods, data-driven DNN techniques aim to establish a model of image quality that is not dependent on manually crafted definitions. This shift towards a more adaptive and self-learning approach enables the model to capture nuanced aspects of image quality that may be challenging for conventional methods to encapsulate.

In a pioneering work [47], a learning-based method has been introduced, employing a comprehensive optimization process that combines feature learning and regression. This method exhibits strong predictive capabilities for assessing the quality of images. In contemporary approaches, the majority of learning-based methods derive visual sensitivity maps from BIQA databases to predict image quality scores [52], [53], [54]. To account for the characteristics of the Human Visual System (HVS), some researchers have developed methods for extracting visual sensitivity maps that align with human perception, yielding favorable outcomes [55], [56]. Additionally, certain learning-based methods have demonstrated outstanding performance in FR-IQA by specifically addressing the perceptual errors associated with inter-image information [57], [58], [59]. These findings underscore the significance of "perceptual loss" between images in relation to human visual perception, playing a pivotal role in IQA.

In [60], an innovative multi-task deep model was introduced to predict image quality by first identifying distortion types and subsequently estimating quality. Another research proposed an integration of features obtained from tasks related to quality score prediction and Natural Scene Statistics (NSS), aiming to enhance IQA [61]. Zhang et al. [62] employed two network models for extracting distinct features, combining them through bilinear pooling to improve representation. These models were pre-trained on tasks such as image classification and distortion type and level identification. Zhu et al. [63] applied deep meta-learning to enhance the model's generalization across various distortions. Su et al. [64] suggested utilizing a self-adaptive hyper network to aggregate both local distortion features and global semantic characteristics. A convolutional neural network-based BIQA model was proposed by Celona and Schettini [65], which approached IQA as a pairwise ranking, regression, and classification problem. Normalization was investigated by Li et al. [66] in the construction of loss functions for IQA, which aided in performance improvement and faster convergence. Despite efforts to generalize across various image distortions and conditions, these techniques struggle to accurately assess image quality on unseen data, leading to potential performance degradation in real-world scenarios.

The advancements have shown that enhancing feature extraction through sophisticated modules can significantly improve the performance of IQA models. One such advancement is the incorporation of Squeeze-and-Excitation (SE) modules [67], [68]. SE modules play a critical role in deep feature responses by explicitly modeling interdependencies between channels. By adaptively rescaling channel-wise feature responses, these modules enable the network to prioritize more informative features, which is essential for accurate IQA. Established architectures like ResNet [69] and variations of VGG [67] can be significantly enhanced by integrating SE modules, leading to superior performance compared to their baselines. For example, in [70], SE is referred to as Channel Attention (CA), where a deep residual network for super-resolution is built using stacked CA modules. This architecture specifically focuses on learning high-frequency input features, resulting in substantial improvements in image quality assessment compared to previous methods.

Numerous BIQA studies rely on fully supervised approaches, demanding extensive annotated image quality data. Given the expense and time associated with image annotation, there has been a shift towards exploring unsupervised methods. Stochastic embedding robustness was proposed by Terhorst et al. [71] as an unsupervised method for facial image quality prediction. Ou et al. [72] used inter-class similarity distribution distance to perform unsupervised facial image quality estimate. Several linear decoders trained in an unsupervised way were used in [73] to capture various image abstraction levels. Curriculum learning was used by the video quality prediction model in [74] to facilitate self-supervised domain adaption. In [75], representation learning was applied to images that underwent quality-preserving alterations, with the distinction of distortion type and degree being treated as an auxiliary task. Even with advancements, unsupervised approaches full potential for evaluating image quality is yet unexplored, especially when it comes to examining data about cross-view representation in unsupervised IQA techniques that are in use currently.

III. TRANSFORMER MODELS FOR FR-IQA

The evaluation of IQA lies at the heart of numerous computer vision applications, influencing the perceptual fidelity and usability of visual content. In the realm of FR-IQA, the challenge is to develop methodologies that accurately gauge the perceived quality of an image in comparison to a high-quality reference. Traditional FR-IQA methods have made significant strides but often struggle to capture the intricate details and complex visual patterns that contribute to human perceptual judgments.

In recent years, transformer-based models has reshaped the landscape of computer vision tasks, demonstrating remarkable success across various domains. The transformative power of transformers in capturing long-range dependencies, understanding spatial context, and facilitating efficient feature extraction has sparked considerable interest in their application to FR-IQA. This section discusses about employing transformers for FR-IQA, exploring how these models transcend the limitations of conventional approaches and elevate the accuracy and robustness of image quality evaluations.

As we navigate through this section, we will unravel the architecture and mechanisms that make transformers well-suited for FR-IQA tasks. By examining key transformer models, understanding their attention mechanisms, and delving into their capacity for spatial context analysis, we aim to provide a comprehensive overview of the strides made in leveraging transformers to enhance the precision and reliability of IQA. From the transfer learning paradigm to fine-tuning strategies tailored for FR-IQA, this exploration aims to shed light on the evolving landscape of IQA propelled by the prowess of transformer models.

In this section, a structured exploration of transformer models for FR-IQA is presented. Table 1 provides a comparison of various transformer models applied to FR-IQA, highlighting their key components, features, and training methodologies. We can mainly categorize these techniques into Hybrid and Multi-scale. The Hybrid category encompasses models that ingeniously combines the strengths of CNNs and transformers. Further, the Multi-Scale category delves into models designed to address quality assessment at different scales, offering a holistic perspective. Within these categories, the techniques are sub-divided based on their approach, featuring CNN-Transformer fusion, Ensemble Approaches, and Joint Training methodologies.

A. CNN-TRANSFORMER INTEGRATION

1) IQT

The Image Quality Transformer (IQT) model was proposed by Cheon et al. [76]. The IQT model comprises a feature extraction backbone, transformer encoder-decoder, and a prediction head. Leveraging a CNN backbone, specifically Inception-Resnet-V2 [81], feature maps from reference and distorted images are extracted. The transformer encoder-decoder, inspired by the standard transformer architecture, processes the embedded feature maps, with the decoder output utilized for quality prediction through an MLP head. The integration of a CNN backbone and transformer encoder-decoder reflects a hybrid approach, capitalizing on the strengths of both architectures for comprehensive IQA.

2) CQT

The CQT (Convolved Quality Transformer) model, introduces a hybrid structure aiming to capitalize on the strengths of both Transformers and pre-trained CNNs [77]. It incorporates three main components: image encoding CNNs, Contextual Quality Transformer (CQT) blocks, and a score predictive MLP head. The model strategically combines long-range dependency extraction with local quality representation, offering a comprehensive approach to IQA. The utilization of CQT blocks enhances the model's ability to comprehend quality differences at both global and local scales, utilizing Multi-Head Self-Attention and convolution layers. The integration of a perceptual pooling function in the final stage reflects a consideration for human visual characteristics, aligning the model's predictions with perceptual qualities.

B. MODULATION NETWORK

1) VTAMIQ

Vision Transformer for Attention Modulated Image Quality (VTAMIQ) presents a holistic approach to FR IQA by leveraging ViT for global representations, introducing context-aware patch sampling, and incorporating attention-based difference modulation [21]. VTAMIQ employs a modified ViT to encode each input image as a single global representation. The model introduces a context-aware patch sampling strategy (CAPS) that leverages human visual attention and biases sampling towards regions with higher perceptual differences. VTAMIQ uses Residual

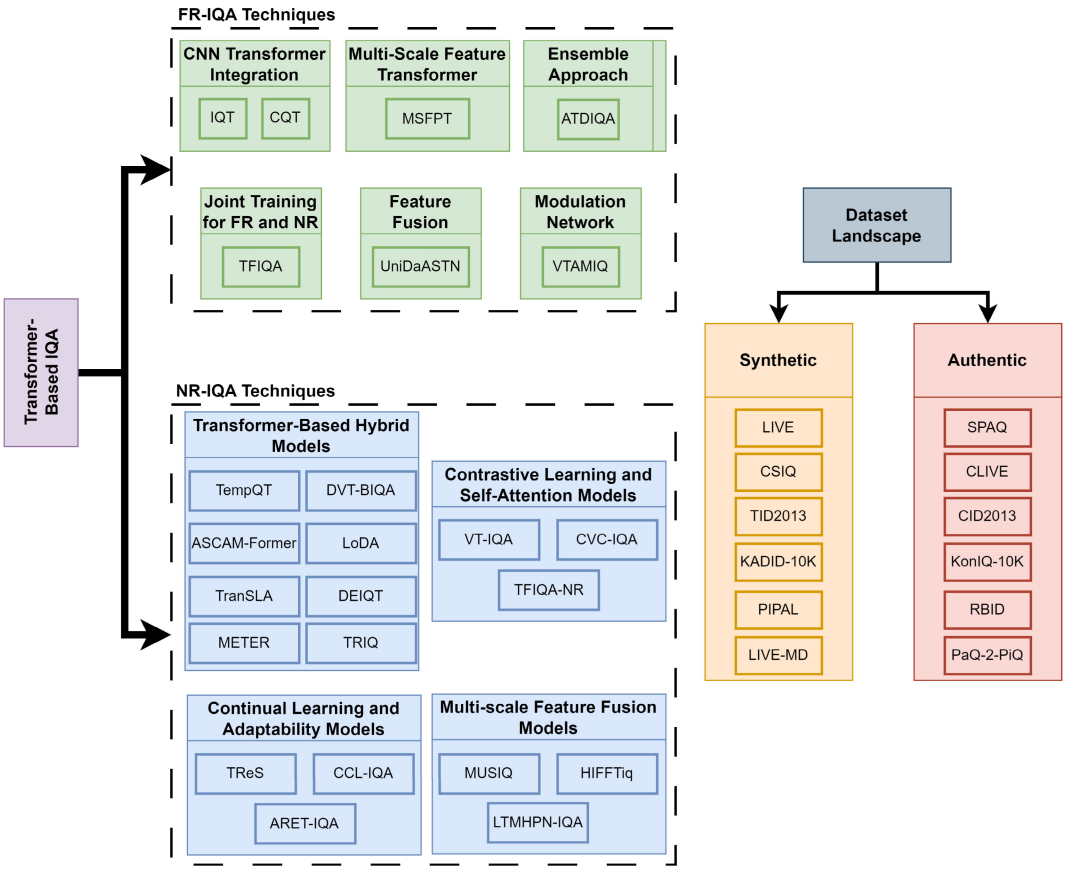

FIGURE 3. Manuscript's bird eye view.

Channel Attention Blocks (RCABs) and Residual Groups (RG) for difference modulation. Channel attention is applied to intelligently adjust the difference vector. The model is trained with Mean Absolute Error (MAE) and benefits from ranking loss, providing a nuanced training signal by emphasizing relative rankings in addition to regression losses.

C. FEATURE FUSION

1) UniDASTN

The proposed UniDASTN (Unifying Dual-Attention and Siamese Transformer Network) model for FR-IQA integrates a Feature Extraction Module, Channel Attention Module, Feature Embedding Module, Spatial Attention Module, and Prediction Head [20]. It leverages a two-branch pre-trained ViT for feature extraction and employs attention mechanisms for channel-wise and spatial information. Utilizing a two-branch pre-trained ViT demonstrates an innovative approach to feature extraction, capturing both low-level and high-level features for a comprehensive quality assessment. The inclusion of both channel and spatial attention mechanisms enhances the model's ability to focus on relevant image features, contributing to IQA. The introduction of a "quality token" in the Feature Embedding Module aligns with the attention to perceptual differences and contributes to a more nuanced understanding of image quality. The model introduces a novel loss function comprising Mean Squared Error (MSE), bidirectional KL divergence, and rank loss components.

D. ENSEMBLE APPROACH

1) ATDIQA

ATDIQA (Attention-based Transformer and CNN Ensemble IQA), is an ensemble of IQT and CNN-based DISTS (Deep Image Structure and Texture Similarity) IQA models [78]. Inspired by the AHIQ model [101], ATDIQA aims to combine the strengths of transformer-based IQA and CNN-based IQA using a feature fusion module. ATDIQA combines IQT and CNN-based DISTS models, utilizing VGG16 [67] to reduce computational costs compared to the deep Resnet50 [69] backbone used in AHIQ. The DISTS-based methods utilize a CNN-based FR-IQA approach, incorporating a feature extraction backbone pre-trained on VGG16 and a weighted summation of texture and structure similarity for each feature level. The IQT component in ATDIQA comprises three levels of feature extraction models (IQT-L, IQT-M, IQT-H) based on the transformer model, inspired by the success of ViT and other transformer-based IQA models.

TABLE 1. Overview of transformer-based models for FR-IQA (The models are categorized based on how they handle features during image quality prediction.)

Category	Model	Components	Loss Functions/Training	Key Features
CNN-Transformer Integration	IQT [76]	CNN (Inception-Resnet-V2), Transformer Encoder-Decoder, MLP Head	MSE	Combines CNN for feature extraction with transformer for quality prediction
	CQT [77]	CNNs, Contextual Quality Transformer Blocks, MLP Head	MAE	Hybrid structure with long-range and local quality representations; perceptual pooling
Feature Fusion	UniDASTN [20]	Feature Extraction Module, Channel Attention, Spatial Attention, Prediction Head	MSE, bidirectional KL divergence, rank loss	Dual-branch ViT, channel and spatial attention, novel loss function (MSE, KL divergence, rank loss)
Modulation Network	VTAMIQ [21]	ViT, Context-Aware Patch Sampling (CAPS), Residual Channel Attention Blocks (RCABs)	MAE, rank loss	Global representation, context-aware patch sampling, difference modulation
Ensemble Approach	ATDIQA [78]	IQT, CNN-based DISTS, Feature Fusion Module	MSE	Ensemble of transformer-based and CNN-based models; uses VGG16 for computational efficiency
Joint Training for FR and NR	TFIQA [79]	CNN Encoder, Transformer Encoder, Attention Module, NR Classifier	MSE	Joint training for FR and NR, combines CNN and transformer features, attention mechanism for fusion
Multi-Scale Feature Transformers	MSFPT [80]	Feature Extraction Block (InceptionNet-V2), Interpolation Block, Transformer Block, Averaging Block	MAE	Multi-scale approach, parallel transformers for different scales, GAN-based distortion handling

E. JOINT TRAINING FOR FR AND NR

1) TFIQA

The TF-IQA (Transformer Features for IQA) model integrates a CNN encoder, transformer encoder, attention module, structure similarity, and NR classifier [79]. It handles both FR and NR datasets, emphasizing joint training for improved IQA. Inspired by transformer advancements, it refines features from CNN layers and combines CNN and transformer strengths using an attention mechanism. The model's unique feature lies in simultaneously training for FR and NR branches, fostering a holistic learning approach. Leveraging VGG16 [67] as the backbone, it reshapes CNN features for the transformer, utilizing attention modules for enhanced fusion. The attentive feature comparison for FR involves a module combining transformer token and CNN channel attention. This model offers a versatile and innovative approach to IQA, addressing local and global information integration.

F. MULTI-SCALE FEATURE TRANSFORMERS

1) MSFPT

The MSFPT (Multi-Scale Features and Parallel Transformer) network, inspired by the NTIRE 2021 challenge winner, is designed for multi-scale IQA [80]. It employs Feature Extraction Block, Interpolation Block, Transformer Block, and Averaging Block. The Feature Extraction Block uses InceptionNet-V2 [102] for feature extraction, followed by an Interpolation Block for consistent feature shapes. The Transformer Block utilizes parallel transformers for different scales, producing quality predictions. Finally, the Averaging Block calculates the overall quality score by averaging scale-specific predictions. This model enhances IQA by considering Generative Adversarial Network (GAN)-based distortions at various scales.

IV. TRANSFORMER MODELS FOR NR-IQA

In the dynamic context of IQA, NR-IQA emerges as a critical domain, particularly relevant due to its unique ability to evaluate image quality without the reliance on reference images. Unlike FR-IQA, which demands a pristine reference for comparison, NR-IQA excels in real-world scenarios where access to such references is often impractical. While traditional approaches have made commendable progress, they often grapple with the intricate details and nuanced patterns crucial for human perceptual judgments.

In this section we discuss the intricate architecture and mechanisms that render transformers well-suited for NR-IQA tasks. By scrutinizing key transformer models, understanding their attention mechanisms, and delving into their capacity for spatial context analysis, our aim is to provide a comprehensive overview of the strides made in leveraging transformers to augment the accuracy and reliability of IQA. From embracing the transfer learning paradigm to tailoring fine-tuning strategies specifically for NR-IQA, this exploration seeks to illuminate the evolving landscape of IQA driven by the inherent capabilities of transformer models. To comprehensively navigate the diverse techniques in this domain, the five distinct categories have been identified, each offering a unique approach to

TABLE 2. Overview of transformer-based models for NR-IQA.

Model	Architecture	Key Features	Techniques
Transformer-Based Hybrid Models			
DVT-BIQA [82]	Dual-Branch Vision Transformer	Dual-scale features, SA-IQA, CA-IQA embeddings	Feature extraction, Branch combination
TempQT [83]	Two-Step NR-IQA, ViT-based	Objective Error Map Model, Perceptual Quality Token	Error map prediction, Quality token fusion
LoDa [84]	Pretrained ViT	ViT and CNN integration, Local distortion features	Cross-attention mechanism, Feature injection
ASCAM-Former [85]	Hybrid CNN and Transformer	Spatial-wise and channel-wise attention, I2PWS scheme	Feature extraction, Attention merging
TranSLA [86]	Saliency-guided Transformer	SaT, LFE, BIM components	Saliency prediction, Local embedding, Interaction boost
DEIQT [87]	Transformer Encoder-Decoder	Quality-aware decoder, Attention-panel mechanism	Self-attention, Quality feature aggregation
METER [88]	Multi-Module Approach	MSFE, DTI, AQP modules	Multi-scale feature extraction, Distortion identification
TRIQ [89]	Hybrid image patching	Patch-based approach, Conv2D-projection	Image patching, Feature map-based strategy
Contrastive Learning and Self-Attention Models			
CVC-IQA [90]	Contrastive Learning Framework	CVC-IM module, PRIs, RMs	Contrastive learning, Self-attention
VT-IQA [91]	Vision-Transformer-based	Global and local distortion handling, Region selection	Feature extraction, Self-attention fusion
TFIQA-NR [92]	ViT and CNN integration	Quality decoder, Channel attention	Feature aggregation, Dual-branch prediction
Continual Learning and Adaptability Models			
CCL-IQA [93]	Contrastive Continual Learning	Continual learning, Attention focal contrastive loss	Task migration, Attention mapping
ARET-IQA [94]	Aspect-Ratio-Embedded Transformer	Ratio-embedded Transformer blocks, (S)W-MSA	Aspect-ratio-preserving self-attention
TReS [95]	Transformers with CNN	Multi-scale features, Relative ranking	CNN feature extraction, Transformer encoder
Multi-scale Feature Fusion Models			
HiFFTiq [96]	Hierarchical Features	EfficientNetV2, Uniform Partition Pooling	Hierarchical feature fusion, MLP head
MUSIQ [97]	Multi-Scale Representation	Multi-scale and aspect ratio-preserving patches	Patch encoding, Scale embedding
LTMHPN-IQA [98]	Lightweight Modules	Overlap Embedding, Lightweight Transformer Blocks	Feature extraction, Multi-Head Prediction
Swin Transformer-Based Approaches			
MSTRIQ [99]	Swin Transformer Backbone	Hierarchical feature fusion, Siamese networks	Dual-loss function, Data augmentation
MANIQA [100]	Multi-dimensional Attention Network	TAB, SSTB, Dual-branch structure	Channel attention, Local and global context

harnessing the power of transformers. These categories encompass Transformer-Based Hybrid Models, Contrastive Learning and Self-Attention Approaches, Continual Learning and Adaptability Models, Multi-Scale Feature Fusion Models, and Swin Transformer-Based Approaches. The transformer-based models for NR-IQA are summarized in Table 2.

A. TRANSFORMER-BASED HYBRID MODELS

1) DVT-BIQA

The model is a dual-branch vision transformer designed for IQA [82]. The architecture comprises feature extraction, a dual-branch vision transformer, and branch combination. Feature extraction is initiated using a ResNet50 [69] backbone network, extracting dual-scale features (S-Feature and L-Feature) crucial for understanding both global and local distortions. The dual-branch vision transformer processes these features, with S- and L-Branches employing transformer encoder layers for scale-specific information aggregation. The model utilizes additional embeddings, SA-IQA and CA-IQA, to enhance scale-specific and content-aware information. The branch combination involves concatenating and processing these embeddings through feed-forward blocks (FFBs) to obtain the final image quality score. The proposed dual-branch vision transformer addresses the complex task of IQA by considering both global and local distortions, acknowledging their importance in real-world scenarios. The use of transformer models for processing dual-scale features demonstrates an effective strategy for capturing intricate patterns at different scales. The incorporation of SA-IQA and CA-IQA embeddings enriches the model's understanding of scale-specific and content-dependent information, contributing to its robustness across various image distortion types.

2) TempQT

The TempQT model is a two-step NR-IQA approach [83]. It comprises an Objective Error Map Model Pre-training step and an Image Score Prediction step. The model employs a Transformer architecture, specifically the ViT, to generate an error map and a perceptual quality token. The decoder aggregates patch embeddings from different layers of the Transformer for error map prediction. Another Transformer is utilized to obtain a perceptual quality token, replacing the class token in the original ViT. This token captures the perceptual impact of each patch on the evaluated image. The two branches of the model, one focusing on distortion information and the other on perceptual attention, are fused. Global Average Pooling is applied to the predicted error map, and the features are combined and regressed into the final subjective quality score. The feature fusion mechanism, combining objective distortion information and perceptual attention, is a strength of the model for comprehensive quality assessment.

3) LoDa

The LoDa framework combines a pretrained ViT with a pretrained CNN to adaptively enhance IQA tasks [84]. The model employs a ViT for global feature extraction and a CNN for multi-scale feature extraction, with a focus on

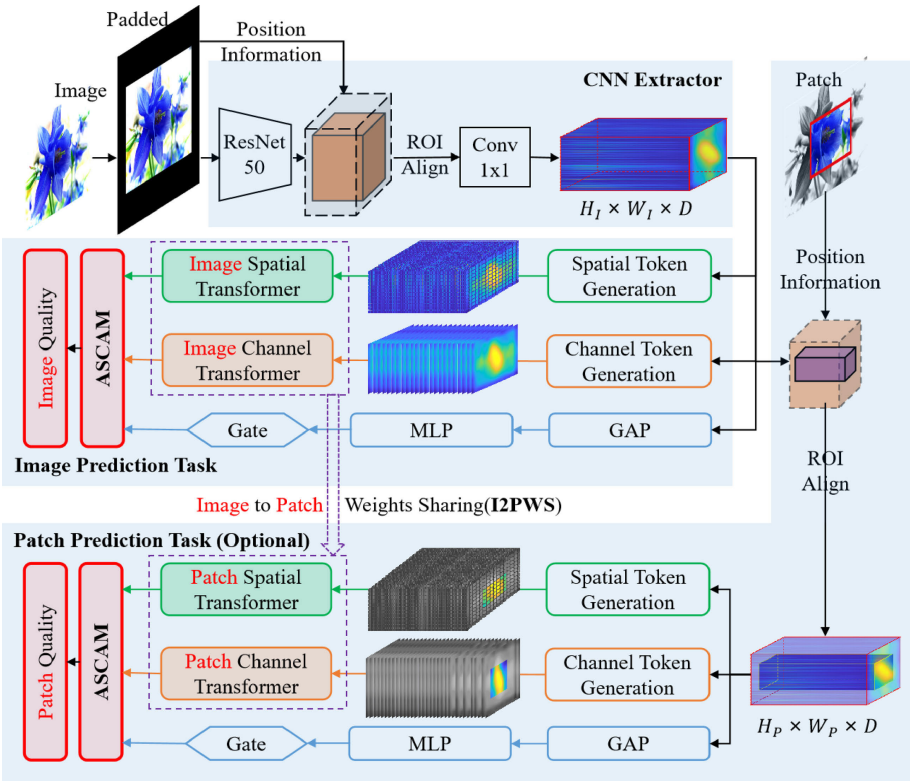

FIGURE 4. Overview of ASCAM-former architecture proposed for IQA [85]. This hybrid architecture, attention mechanisms, and I2PWS scheme collectively contribute to the model's technical robustness for IQA tasks, particularly in scenarios with multi-scale quality labels. *Source:* Image adapted from Ma et al. [85].

local distortion features. The Local Distortion Extractor and Injector components efficiently integrate local distortion information into the ViT framework using a cross-attention mechanism. During adaptation, specific components are trainable, while the weights of the pretrained ViT encoder and CNN are frozen. The integration of pretrained ViT and CNN addresses both non-local and local dependencies in image features, offering a comprehensive approach to IQA. The use of a cross-attention mechanism for injecting local distortion features demonstrates a thoughtful strategy to ensure effective integration without overloading parameters. The freezing of pretrained weights during adaptation helps leverage the knowledge gained from large-scale datasets without losing computational efficiency. The focus on local distortion features is a valuable addition, considering the importance of both global and local information in image quality evaluation.

4) ASCAM-Former

The ASCAM-Former model presented in Figure 4 adopts a hybrid architecture, incorporating both CNNs and Transformers to enhance image quality prediction [85]. The initial stage involves a ResNet50-based CNN Extractor [69], which efficiently captures local features and preserves intricate details from input images. This combination of CNNs and Transformers allows the model to leverage the strengths of both architectures. The architecture branches into two separate paths for spatial-wise and channel-wise attention learning as shown in Figure 4. The spatial-wise attention branch focuses on spatial features, while the channel-wise attention branch extracts channel-specific characteristics. This bifurcation ensures a nuanced understanding of the spatial and channel dimensions of the input data. A distinctive aspect of the model is the attention merging branch. This branch dynamically controls the contributions of spatial and channel branches through the generation of customized gate embeddings. This adaptability is crucial for effectively balancing the impact of spatial and channel information on the final image quality prediction, leading to improved performance in diverse scenarios.

Moreover, the ASCAM-Former introduces the I2PWS (Image-to-Patch Weight Sharing) scheme, designed to facilitate end-to-end training on datasets with both image and patch labels. This scheme involves sharing transformer weights between image and patch prediction tasks, even though their configurations differ. This approach ensures knowledge transfer between global (image-level) and local (patch-level) quality prediction tasks, enhancing the overall learning process. The ASCAM-Former's hybrid architecture,

attention mechanisms, and the I2PWS scheme collectively contribute to its technical robustness, making it well-suited for IQA tasks, particularly in scenarios where datasets contain multi-scale quality labels.

5) TranSLA

The framework aims to enhance feature learning in Transformer-based NR-IQA by integrating different-level information [86]. The architecture consists of three main components which are Saliency-Guided Transformer Module (SaT), Local Feature Embedding Module (LFE) and Boosting Interaction Module (BIM). The SaT module incorporates a Saliency Prediction Network (SPN) to guide the Transformer's attention to the Region of Interest (RoI) in the image. The SPN generates a saliency map, and the Transformer processes raw features using this map to focus on relevant regions. LFE introduces local embedding to improve the Transformer's performance. It utilizes gradient maps, which explicitly represent high-frequency information in the image. The gradient map is combined with the distorted image to provide additional local information to the Transformer. BIM is designed to enhance information interaction in Transformer. It splits the output of each block into patch tokens and a class token. To boost interaction, it calculates the weight of each layer's class token using Kullback-Leibler divergence and applies it as a loss during training. The TranSLA framework with Local Embedding demonstrates its effectiveness through innovative modules that enhance attention, consider local features, and promote effective information interaction.

6) DEIQT

The Data-Efficient Image Quality Transformer (DEIQT) operates on the principle of enhancing BIQA through a Transformer encoder-decoder framework [87]. DEIQT leverages the Transformer's self-attention mechanism to capture both local and non-local dependencies, facilitating the aggregation of comprehensive perceptual features from input image patches. It introduces a quality-aware decoder to interpret the CLS token, enhancing the model's ability to extract quality-related information. The attention-panel mechanism further reduces prediction uncertainty by considering image quality from various perspectives, akin to subjective evaluations by multiple participants. DEIQT's working principle integrates these components, aiming to efficiently learn and represent image quality features, ultimately improving the accuracy and robustness of BIQA predictions.

7) METER

METER employs a multi-module approach, featuring a Multi-Scale Semantic Feature Extraction (MSFE) backbone, a Distortion Type Identification (DTI) module, and an Adaptive Quality Prediction (AQP) module [88]. MSFE extracts multi-level local and global features, DTI identifies distortion types, and AQP predicts image quality by adapting to various content and distortion types. The backbone of MSFE utilizes an Efficient Vision Transformer (EVT) for low-level feature extraction and stages for multi-scale feature maps. While DTI module predicts distortion types using multi-scale semantic features from MSFE, employing fully connected layers and a softmax function. Further, the AQP module diversifies image content and distortion types, utilizing an adaptive fully-connected (AFC) layer with learned parameters. METER introduces a unique combination of multi-scale feature extraction, distortion type identification, and adaptive quality prediction. Its novel use of an EVT backbone, LDA blocks, and AFC layers enables effective handling of diverse image characteristics. The fusion of distortion-type probabilities and quality scores enhances its versatility, making METER a robust model in perceptual IQA.

8) TRIQ

TRIQ employs a hybrid approach, combining image patching and a feature map-based strategy inspired by ViT [89]. It projects image patches or feature maps using a Conv2D-projection layer into a transformer encoder. TRIQ adapts to diverse image resolutions without resizing, using max-pooling for memory efficiency. The Transformer encoder produces an aggregated vector fed into an MLP head for quality prediction. Quality distribution over grade-to-vote is predicted using Softmax activation. TRIQ stands out as a novel model by seamlessly integrating image patching and feature map-based approaches within a Transformer framework. Its adaptability to diverse resolutions without resizing ensures practicality in real-world scenarios. The quality distribution prediction through an MLP head enhances robustness, making TRIQ a versatile and effective model for IQA.

B. CONTRASTIVE LEARNING AND SELF-ATTENTION MODELS

1) CVC-IQA

The model utilizes a contrastive learning framework with multi-views, featuring a Cross-View Consistent Information Mining (CVC-IM) module, pseudo-reference images (PRI), Residue Maps (RM), and a Transformer-based self-attention mechanism [90]. The CVC-IM is designed for self-supervised extraction of mutual information among different views while employing contrastive learning to distinguish between consistent and inconsistent sample pairs. The representation learning utilizes PRIs and RMs of different distortion types to enhance feature representations. Contrastive learning on PRI and RM views facilitates learning complementary information for quality-aware fidelity. Transformer is used for self-attention across feature embeddings. The transformer architecture combines LAB feature extractor and PRI-RM feature extractor for comprehensive feature learning. The model presents a well-thought-out architecture that leverages contrastive learning and self-attention for effective IQA. The inclusion of diverse distortion types

and multi-view learning contributes to its generalization capabilities.

2) VT-IQA

The VT-IQA model is a Vision-Transformer-based BIQA model designed for simultaneously considering both global content and multi-locational local distortions in images [91]. The architecture comprises Input Module, Feature Extraction Module, Fusion Module, and Regression Module. The input module generates two images, Ic for global content and Id for multi-locational local distortions. It uitilizes a region selection strategy based on sensitivity analysis to choose representative patches for Id. The feature extraction module employs a ViT to extract global content feature (fc) from Ic and local distortion feature (fd) from Id while preserving position information. The feature fusion model integrates local and global features using a self-attention (SA) based fusion module. At the end a regressor consisting of fully connected layers is used to predict the image quality score from the fused feature. The VT-IQA model presents a comprehensive approach to BIQA, effectively integrating global and local information for accurate predictions. The use of Vision-Transformer and self-attention mechanisms enhances the model's ability to capture intricate details in images.

3) TFIQA-NR

The proposed TFIQA-NR model integrates a ViT as the visual backbone with a CNN channel attention-based quality decoder [92]. The ViT encoder captures global contextual information by tokenizing input images and using self-attention mechanisms. Feature aggregation combines information from different transformer stages, and the quality decoder, featuring residual channel attention blocks, refines the features for improved quality prediction. The dual-branch quality predictor employs both spatial attention and linear layers. The model demonstrates potential strengths in capturing global relationships and refining features for improved IQA.

C. CONTINUAL LEARNING AND ADAPTABILITY MODELS

1) CCL-IQA

The method, Contrastive Continual Learning for IQA (CCL-IQA), addresses the robustness of migration to unknown image distortion types in the context of transformer-based BIQA [93]. It introduces a continual learning approach to reduce catastrophic forgetting during the migration of new tasks, enhancing the model's overall robustness. The model architecture involves a Transformer model for image feature extraction and a novel attention focal contrastive loss to preserve prior knowledge. Attention mapping in the Transformer is adapted by computing the similarity between self-queries and learnable external keys, biased towards previous task information. The attention focal contrastive loss is formulated to compare attentional focus vectors between the new task and old tasks, effectively accumulating prior knowledge. The total loss used in the BIQA approach is a combination of attention focal contrastive loss and BIQA loss functions. The framework is designed to enhance the model's adaptability to new tasks while preserving knowledge from previous tasks, thereby improving the overall robustness of the IQA model.

2) ARET-IQA

The Aspect-Ratio-Embedded Transformer (ARET) for IQA, is designed to effectively utilize original aspect ratios in the self-attention computation of the Transformer model for comprehensive IQA [94]. The architecture is based on the Swin Transformer model. The model consists of multiple ratio-embedded Transformer blocks, each embedding the original aspect ratios into the computation. Images are split into non-overlapping patches, and both patches and original aspect ratios are embedded into the Transformer blocks. Unique attention mechanism, namely Aspect-Ratio-Preserving Window-based Multihead Self-Attention ((S)W-MSA), is proposed in each ratio-embedded Transformer block. It adjusts self-attention computation based on aspect ratios. Global average pooling is used to obtain output features from multiple Transformer blocks. A prediction head, consisting of a two-layer MLP, maps the combined features to predict perceptual image quality.

3) TReS

The TReS (Transformers, Relative ranking, and Self-consistency) method leverages a combination of CNNs and Transformers for feature extraction and attention-based feature computation [95]. The model utilizes a CNN with learnable parameters to extract features from different layers. The features from each block of the CNN are normalized, pooled using Euclidean norm, and subjected to dropout layers. Multi-scale features from different layers are concatenated to form the final feature representation. To capture both low- and high-level features, the transformer's encoder architecture is employed. The features from different layers of the CNN are treated as a sequence and sent through the transformer encoder. The self-attention mechanism in transformers helps compute non-local representations, capturing complex relationships across features. Positional encoding is introduced to make the model aware of feature positions. Fully connected (FC) layers are used to fuse features obtained from local (CNN) and non-local (Transformer) operators. The model considers the relative ranking relation between samples within each batch to enhance correlation. Triplet loss is utilized to enforce relative ranking, addressing extreme cases within the batch. The model leverages self-consistency by enforcing equivariant transformations on the input images. The combination of relative ranking and self-consistency contributes to the model's ability to learn robust representations for IQA.

D. MULTI-SCALE FEATURE FUSION MODELS

1) HiFFTiq

HiFFTiq aims to address the challenge of perceiving image quality by leveraging hierarchical features and a Transformer-based fusion approach [96]. The model utilizes EfficientNetV2 as the backbone for hierarchical feature extraction (HFP). HFP is introduced to transform input features of different sizes and dimensions into sequences of fixed length and dimension. Uniform Partition Pooling (UPP) is proposed as an alternative to Sliding Window Pooling (SWP) to assess quality uniformly in different areas. The hierarchical features are fused using Transformer encoders. Shallow Transformer encoders capture image quality for each input sequence, and a deep Transformer encoder is used to fuse hierarchical features. An additional "Image Quality token" is introduced to represent the entire hierarchical input. The fused features are then fed into a MLP head for IQA.

2) MUSIQ

The MUSIQ model is designed to handle inputs with arbitrary aspect ratios and resolutions, providing a multi-scale representation for effective IQA [97]. The model generates a multi-scale representation of the input image, including the native resolution image and its aspect ratio-preserving resized variants. A patch encoding module, consisting of a 5-layer ResNet with a fully connected layer, is used to encode each patch into a D-dimensional embedding. Square patches of size P are extracted from each image at different scales. The patches are encoded into embeddings using the patch encoding module. Hash-Based 2D Spatial Embedding (HSE) is introduced to encode the 2D spatial position for each patch. Patches located at different scales are mapped to the same grid to achieve spatial alignment across different images. Scale Embedding (SCE) is introduced to distinguish information coming from different scales in the multi-scale representation. During pre-training, the model is trained on large datasets with random cropping as an augmentation. Fine-tuning is performed on IQA tasks without resizing or cropping the input image.

3) LTMHPN-IQA

The LTMHPN-IQA method consists of two lightweight modules, the MFE (Module of Feature Extraction) and the MMHP (Module of Multi-Head Prediction) [98]. MFE takes a distorted image as input and employs Overlap Embedding, which involves using a convolutional kernel for overlapping patch embedding. It utilizes six Lightweight Transformer Blocks (LTBs) for feature extraction at different scales. The LTBs consists of a Lightweight Attention Block (LAB) and a Depth-Wise Convolution Block (DWCB). LAB focuses on channel attention using Point Convolution (PConV) and DWConV operations. DWCB involves doubling the channel dimension, performing feature aggregation with DWConV, and then halving the channel dimension. Point Convolution (PConV) and Depth-Wise Convolution (DWConv) are utilized for spatial information aggregation. The Module of Multi-Head Prediction (MMHP) take features from MFE and employs three Weighted Prediction Blocks (WPBs) to measure the importance of different elements of input features at the same scale. While MPB consists of two branches feature branch (PConV and ReLU) for aggregating features and weight branch (PConV and Sigmoid) for learning feature element weights. Lastly, the framework utilizes an FC layer for aggregating features at different scales by adding adaptive weights.

E. SWIN TRANSFORMER-BASED APPROACHES

1) MSTRIQ

MSTRIQ utilizes a Swin Transformer backbone for IQA [99]. Cropped patches from resized images are processed through Swin Transformer blocks, and features from multiple stages are fused to capture hierarchical information. A fused feature map is obtained and mapped to image quality score using an MLP header. Siamese networks enable pair-wise training, and a dual-loss function combines rank and regression loss for model optimization. Data augmentation techniques enhance robustness. Euclidean loss is used for regression, measuring the absolute distance between predicted and ground truth scores. Pair-wise ranking loss, addressing relative rankings, is employed using an exponential form. The proposed model introduces a novel approach by fusing features from different stages of a Swin Transformer for comprehensive hierarchical information capture. The combination of Siamese networks, pair-wise training, and a dual-loss function (rank and regression) enhances the model's ability to handle both absolute and relative quality assessments. The utilization of data augmentation techniques further contributes to the robustness of the model, making it a promising solution for NR-IQA tasks.

2) MANIQA

The MANIQA (Multi-dimensional Attention Network for IQA) model employs a novel approach for IQA [100]. It incorporates a Transposed Attention Block (TAB) that applies self-attention across channel dimensions to capture global context. The TAB rearranges channel weights based on their importance to perceptual quality scores, providing an effective way to utilize information from different channels. Additionally, it features a Scale Swin Transformer Block (SSTB) to enhance local information. The SSTB incorporates Swin Transformer Layers (STL) and a convolutional layer to encode features from different stages. The model utilizes a dual-branch structure for patch-weighted quality prediction, balancing scoring and weighting branches to prevent overfitting. MANIQA stands out as a novel model in IQA by effectively incorporating attention mechanisms across channel dimensions and integrating Swin Transformer architecture. The combination of these components contributes to its ability to capture both global and local context, making it a promising approach for accurate image quality prediction.

V. DATASET LANDSCAPE IN IQA FRAMEWORKS

In order to comprehensively evaluate the performance of Transformer architectures in perceptual IQA, a diverse set of datasets has been curated, each offering unique challenges and characteristics. These datasets play a crucial role in training and testing models, enabling researchers to explore the robustness and generalization capabilities of their proposed architectures. In this section, we present an overview of key datasets, each contributing unique challenges. These datasets, sourced from various domains, enable a comprehensive exploration of how models respond to different distortions and real-world scenarios. Through detailed analyses of perceptual quality across a spectrum of images, this compilation serves as a critical reference for understanding the nuanced interactions between Transformer architectures and the complexities of IQA. We categorize datasets into two distinct subsections: synthetic datasets, designed for controlled environments, and authentic datasets, representing real-world scenarios. This division enables a comprehensive exploration of IQA models across varying data sources and complexities.

A. SYNTHETIC DATASET

Synthetic datasets plays an important role in advancing IQA research by providing controlled environments with precisely introduced distortions. These datasets facilitate systematic experimentation, allowing researchers to evaluate the robustness and efficacy of IQA models under specific distortion types and levels. In this subsection, we discuss synthetic datasets that emulate diverse distortions, enabling a detailed analysis of model performance and generalization.

1) LIVE

The LIVE dataset [103] comprises 779 synthetically distorted images, derived from 29 reference images. The dataset is designed for IQA and includes images distorted using various techniques, encompassing the following distortion types: JPEG compression (JPEG), JPEG2000 compression (JP2K), white Gaussian noise (WN), Gaussian blur (GB), and fast fading error (FF). For each image in the dataset, perceptual image quality is measured using the Difference Mean Opinion Score (DMOS) metric, ranging from 0 to 100. The dataset provides a diverse set of images with different types of distortions, making it valuable for evaluating the performance of IQA models across various distortion scenarios.

2) CSIQ

The CSIQ dataset [104] comprises 866 synthetically distorted images derived from 30 reference images, encompassing six distinct distortion types: JPEG compression (JPEG), JPEG2000 compression (JP2K), white Gaussian noise (WN), Gaussian blur (GB), contrast change (CC), and pink noise (PN). Similar to the LIVE dataset, CSIQ utilizes the DMOS as the perceptual image quality measure, with scores ranging from 0 to 1. This dataset serves as a valuable resource for evaluating IQA models, offering a diverse set of distortions to assess the robustness and generalization capabilities of such models.

3) TID2013

The TID2013 dataset [105] is a synthetically distorted dataset containing 25 reference images and 3000 distorted images. TID2013 offers a diverse range of distortions, primarily comprising traditional types such as quantization noise, Gaussian blur, compression artifacts, and distortions caused by image denoising, among others. Notably, the MOSs for TID2013 are obtained in controlled laboratory environments, providing a standardized setting for assessing the perceptual quality of distorted images. With its carefully curated distortions and subjective quality annotations, TID2013 serves as a valuable benchmark for evaluating IQA models, particularly in the context of traditional distortions encountered in various real-world scenarios.

4) KADID-10K

The KADID-10k dataset, as introduced by [106], comprises a collection of 10,125 images deliberately distorted through synthetic means. This dataset is derived from 81 reference images and encompasses 25 distinct distortion types, each applied at five different degradation levels.

5) PIPAL

The PIPAL dataset [107], introduced by the NTIRE 2022 Perceptual IQA Challenge [108], distinguishes itself with its unique characteristics, specifically in terms of image distortions. This dataset encompasses a substantial collection of 250 reference images and 29,000 distorted images, strategically generated to represent a diverse array of distortions originating from 40 different types. Notably, PIPAL incorporates distortions arising from both traditional image processing techniques and those generated by state-of-the-art GAN-based image enhancement algorithms.

The dataset's composition includes outputs from various GAN-based algorithms, contributing to its comprehensive representation of the evolving landscape of image enhancement technologies. This distinctive feature positions PIPAL as an invaluable resource for evaluating IQA models within the context of contemporary advancements in image processing. The dataset further divides into a validation set, comprising 25 reference images and 1,650 distortion images, and a testing set, which also consists of 25 reference images paired with 1,650 distortion images. This meticulous division ensures a robust evaluation framework, encompassing a wide spectrum of scenarios and challenges for perceptual quality assessment tasks.

6) LIVE MD

The LIVE MD dataset [109], is a compilation of 450 images generated from 15 reference images, incorporating two distinct multiple distortion types. The distortions include

a sequence of blur followed by JPEG compression and another sequence of blur followed by white Gaussian noise (WN). The perceptual quality of images in the LIVE MD dataset is quantified using the DMOS metric, providing a comprehensive measure of perceived image quality. The DMOS values range within [0, 100], allowing for a nuanced evaluation of the impact of multiple distortions on perceptual quality.

B. AUTHENTIC DATASET

Authentic datasets capture the complexity and diversity of real-world scenarios, presenting a challenge for IQA models. Unlike synthetic datasets, authentic datasets incorporate the inherent variability and unpredictability of images encountered in everyday situations. This subsection explores datasets sourced from authentic scenarios, shedding light on the adaptability and real-world applicability of IQA models in the face of diverse and dynamic image conditions.

1) KonIQ-10K

The KonIQ-10k dataset is a collection of up to 10,073 synthetically distorted images carefully selected from the YFCC100M dataset [110]. The subjective labels for quality assessment are obtained through evaluations by 1,459 crowd workers [111]. Each image in the dataset is assessed by more than 120 raters, accumulating a total of 1.2 million ratings. The labels are meticulously gathered to ensure a comprehensive evaluation of image quality.

2) SPAQ

The SPAQ dataset is a collection of 11,125 images captured using 66 different smartphones [112]. This recently released dataset is notable for its extensive annotations, encompassing various perceptual quality attributes. The provided annotations include scores for contrast, brightness, sharpness, noisiness, colorfulness, and an overall Mean Opinion Score (MOS) indicating the overall image quality. Additionally, the dataset provides information about the scene captured in each image and the image file format.

3) CLIVE

The CLIVE dataset [113] stands out as the pioneering in-the-wild image quality dataset. Comprising 1162 images, this dataset has garnered over 350,000 opinion scores from a vast pool of more than 8100 human observers. Each image in LIVEC is meticulously evaluated by a substantial number of human assessors, contributing to a comprehensive and diverse set of perceptual quality ratings.

4) CID2013

The CID2013 dataset [114] represents an authentic IQA dataset carefully curated in a controlled laboratory environment. This dataset comprises a total of 480 images, which are categorized across eight specific scenes. For each image within the CID2013 dataset, an average of 31 subjective ratings is available, reflecting the diversity of human opinions on perceived image quality.

5) RBID

The RBID (Real Blur Image Dataset) [115] is designed to address authentic blur distortions encountered in real-world scenarios. This dataset encompasses a diverse collection of 585 images, each exhibiting different types of blur distortions. The dataset deliberately includes images with simple motion blur, images with complex motion blur, and images with out-of-focus blur.

6) PaQ-2-PiQ

Representing one of the most extensive IQA datasets to date, PaQ-2-PiQ [116] encompasses a staggering 40,000 real-world images along with 120,000 cropped patches. Each image and associated patch within the dataset is annotated with a MOS, providing a robust measure of perceived quality. Notably, PaQ-2-PiQ stands out by offering quality labels not only for entire images but also for individual patches with varying proportions. This inclusion of fine-grained quality assessments enhances the dataset's utility, allowing for a nuanced exploration of image quality dynamics at both holistic and localized levels.

VI. COMPARATIVE ANALYSIS OF TRANSFORMER-BASED MODELS

A. EVALUATION METRICS AND CRITERIA

The evaluation of AI models in perceptual IQA revolves around two key metrics recommended by the Video Quality Experts Group (VQEG): Spearman Rank Order Correlation Coefficient (SROCC) and Pearson Linear Correlation Coefficient (PLCC) [117]. These metrics provide valuable insights into the model's ability to capture the ranking and linearity of image quality, respectively.

1) NONLINEAR MAPPING FUNCTION

Before delving into the evaluation metrics, a nonlinear mapping function is employed to enhance the comparability of predicted quality scores. A commonly used monotonic logistic function is employed to map computed quality scores o to p as follows:

$$p = \beta_1 \left(0.5 - \frac{1}{1 + e^{\beta_2 (o - \beta_3)}}\right) + \beta_4 o + \beta_5, \tag{6}$$

where $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5$ are parameters derived during the training process.

2) EVALUATION CRITERIA

Following the nonlinear mapping, the performance of Transformer models is quantified using SROCC and PLCC. These metrics assess the model's ability to capture the monotonic relationship and linear correlation with subjective quality scores.

- **Spearman Rank Order Correlation Coefficient (SROCC):**

$$\text{SROCC} = 1 - \frac{6}{N(N^2-1)} \sum_{n=1}^{N} d_i^2, \tag{7}$$

where d_i represents the difference between the i-th image's ranks in subjective and objective evaluations, and N is the number of testing images.

- **Pearson Linear Correlation Coefficient (PLCC):**

$$\text{PLCC} = \frac{\sum_{i=1}^{N}(p_i - \bar{p})(s_i - \bar{s})}{\sqrt{\sum_{i=1}^{N}(p_i - \bar{p})^2 \sum_{i=1}^{N}(s_i - \bar{s})^2}}, \tag{8}$$

where s_i and p_i indicate the i-th image's subjective score and converted objective rating after nonlinear mapping, $\bar{s}$ and $\bar{p}$ are the mean of all s_i and p_i.

Focusing on SROCC and PLCC aligns with the prevalent evaluation practices in the literature for Transformer-based frameworks in perceptual IQA [117]. These metrics collectively gauge the model's efficacy in capturing both ranking and linear relationships with subjective quality scores.

B. PERFORMANCE ANALYSIS OF TRANSFORMER-BASED MODELS

In this section, we conduct a comparative analysis of Transformer-based models for perceptual IQA. Each model is evaluated across diverse datasets to assess its generalization capabilities and performance under different distortions. We utilize the reported results from the original papers for each Transformer-based model, focusing on SROCC and PLCC. These metrics offer insights into the model's ability to capture ranking and linear relationships with subjective quality scores.

1) COMPARISON RESULTS FOR FR-IQA TECHNIQUES

The performance comparison across various FR-IQA techniques on distinct datasets reveals insights into the strengths and constraints of each model. The evaluation metrics, showcase the model's ability to capture perceptual quality across varied image distortions. The results are displayed in Table 3, which summarizes performance metrics reported in the literature. These results are sourced from various studies and reflect benchmarks established by previous research.

The UniDASTN model consistently emerges as a top performer, demonstrating remarkable SROCC and PLCC scores across multiple datasets, including LIVE, CSIQ, TID2013, and PIPAL. For instance, the UniDASTN model achievs SROCC scores of 0.976 on LIVE, 0.942 on CSIQ, and 0.877 on TID2013. These scores are notably higher compared to several other models. For example, DVT-IQA reports SROCC values of 0.976 on LIVE but lower values of 0.877 on TID2013. The integration of a two-branch pre-trained ViT, attention mechanisms, and a unique "quality token" in UniDASTN model contributes to its robust performance. The attention to both channel-wise and spatial information, coupled with an innovative loss function, positions UniDASTN as a comprehensive and effective FR-IQA solution. UniDASTN's success on TID2013 suggests that its dual-attention mechanism, capable of discerning both local and global distortions, contributes to its effectiveness across varied scenarios. In contrast, the VTAMIQ model excels in scenarios dominated by global distortions, with a notable SROCC score of 0.976 on LIVE. However, its performance on TID2013 in terms of SROCC is comparatively lower at 0.950.

On the other hand, ATDIQA presents a unique ensemble approach, combining transformer-based IQT and CNN-based DISTS models. This fusion of two distinct architectures provides a diverse set of features for assessment. However, its performance seems to vary across datasets for example, it achieves an SROCC score of 0.975 on LIVE but only 0.931 on KADID-10k, suggesting that the effectiveness of ensemble models may depend on the specific characteristics of the distortions present in the dataset.

The VTAMIQ model, leveraging a modified ViT for global representations and introducing context-aware patch sampling, excels in capturing perceptual differences, as seen in its SROCC score of 0.970 on LIVE. Its attention to regions with higher perceptual disparities, coupled with the use of RCABs and RG, contributes to its success.

It's noteworthy that the results for the PIPAL dataset specifically pertain to the validation set. It's a common practice in the literature to compute results on the validation set of PIPAL, given its distinct composition of distortions induced by both traditional and GAN-based image enhancement algorithms. The emergence of contemporary distortions introduced by GAN-based algorithms, as represented in the PIPAL dataset, adds another layer of complexity to the evaluation. Models like UniDASTN demonstrate resilience against these novel distortions, with competitive performance metrics such as a PLCC score of 0.831 PLCC. This indicates their ability to adapt and generalize well in the dynamic landscape of image processing technologies.

2) COMPARISON RESULTS FOR NR-IQA TECHNIQUES ON SYNTHETIC DATASETS

Table 4 presents the performance comparison of various NR-IQA methods across multiple synthetic datasets, including LIVE, CSIQ, TID2013, KADID-10k, and PIPAL. The performance metrics shown are based on results reported in existing literature. The evaluation metrics include SROCC and PLCC. This comparison provides insights into the strengths and weaknesses of each NR-IQA method across different synthetic datasets. The NR-IQA methods evaluated on synthetic datasets showcase varying degrees of success in capturing and assessing image quality. Several NR-IQA methods, such as DVT-IQA, TempQT, and TFIQA-NR, demonstrate consistent and high performance across various synthetic datasets.

The majority of the methods exhibit robust performance across different distortion types within each dataset.

TABLE 3. Performance Comparison of FR-IQA methods on different datasets. (The bold values indicate the highest performance, while the underlined values represent the second-highest performance.)

Method	LIVE		CSIQ		TID2013		KADID-10k		PIPAL	
	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
IQT	0.970	-	0.943	-	0.899	-	-		0.820	0.840
CQT	0.958	0.951	0.964	0.962	0.888	0.896	-	-	-	-
ATDIQA	0.931	0.931	-	-	0.795	0.837	-	-	-	-
TFIQA	0.958	0.947	0.938	0.941	0.832	858	-	-	-	-
MSFPT	0.977	0.972			0.92	0.929	0.883	0.888	0.81	0.788
VTAMIQ	0.982	0.984	0.979	0.982	0.950	0.954	**0.963**	**0.964**	-	-
AHIQ	0.984	**0.989**	0.975	0.978	0.962	0.968	-	-	**0.835**	**0.845**
UniDASTN	**0.987**	0.989	**0.983**	**0.987**	**0.974**	**0.977**	-	-	0.829	0.831

TABLE 4. Performance comparison of methods on synthetic datasets. (The bold values indicate the highest performance, while the underlined values represent the second-highest performance.)

Method	LIVE		CSIQ		TID2013		KADID-10k		PIPAL		LIVEMD	
	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
TempQT	0.976	0.977	0.950	0.960	0.883	0.906	-	-	-	-	-	-
LoDa	0.975	0.979	-	-	0.869	0.901	0.931	0.936	-	-	-	-
ASCAM-Former	-	-	-	-	0.883	-	0.897	0.901	-	-	-	-
CCL-IQA	0.9702	-	-	-	0.8845	-	0.8092	-	-	-	-	-
TReS	0.969	0.968	0.922	0.942	0.863	0.883	0.859	0.858	-	-	-	-
LTMHPN-IQA	0.943	0.952	0.908	0.934	0.900	0.910	0.944	0.946	-	-	-	-
MSTRIQ	-	-	-	-	0.882	0.895	-	-	**0.954**	**0.946**	-	-
DEIQT	0.980	0.982	0.946	0.963	0.892	0.908	0.889	0.887	-	-	-	-
TFIQA-NR	0.981	0.982	0.958	0.969	0.918	0.923	0.937	0.942	-	-	-	-
METER	0.977	0.978	**0.982**	**0.982**	0.900	0.910	0.944	0.946	-	-	-	-
DVT-IQA	0.976	0.976	0.942	0.952	0.877	0.894	**0.970**	**0.971**	-	-	**0.935**	**0.935**
MANIQA	**0.982**	**0.983**	0.961	0.968	**0.937**	**0.943**	0.944	0.946	-	-	-	-

DVT-IQA and TempQT stand out with high performance across multiple datasets. For instance, DVT-IQA achieves an SROCC of 0.976 on LIVE and 0.942 on CSIQ, while TempQT achieves 0.976 and 0.950, respectively. This consistency highlights the robustness of these methods in handling a variety of synthetic distortions. On the other hand, TFIQA-NR demonstrates exceptional performance on the LIVE dataset with SROCC and PLCC scores of 0.981 and 0.982, respectively, suggesting its effectiveness in capturing perceptual quality in synthetically distorted images.

The dual-branch Vision Transformer employed by DVT-IQA and the transposed attention block in MANIQA allow these models to effectively capture both global and local distortions. For instance, MANIQA shows a high PLCC score of 0.983 on LIVE and a competitive 0.944 on KADID-10k, demonstrating its capability to adapt to diverse distortion types. Attention mechanisms, such as those implemented in TempQT and TFIQA-NR, further enhance these models' stability and prevent overfitting, leading to their strong performance across datasets.

Techniques such as the dual-branch vision transformer employed by DVT-IQA and the transposed attention block used by MANIQA enable these models to adeptly capture both global and local distortions, contributing to their adaptability to a wide array of synthetic distortions. Attention mechanisms, such as rearranging channel weights and computing non-local representations, enhance the models' ability to understand complex relationships within image features. The combination of scoring and weighting branches, as seen in TempQT and TFIQA-NR, contributes to the models' stability and prevents overfitting. The attention to patch-weighted quality prediction, as in MANIQA, ensure a nuanced understanding of the impact of individual image patches on overall perceptual quality. For instance, MANIQA shows a high PLCC score of 0.983 on LIVE and a competitive 0.944 on KADID-10k, demonstrating its capability to adapt to diverse distortion types. The model collectively showcase a harmonious integration of global and local information, facilitating accurate predictions of image quality across diverse synthetic distortions.

The proficiency of these models lies in their adept balancing of scale-specific information and content-aware features. Transformer-based architectures, with their inherent self-attention mechanisms, showcase effectiveness in cap-

turing intricate patterns at different scales. The integration of attention mechanisms, including rearranging channel weights based on importance and computing non-local representations, enhances the models' ability to comprehend complex relationships within image features. These techniques collectively allow the models to seamlessly integrate global and local information, facilitating accurate predictions of image quality across a spectrum of synthetic distortions.

While NR-IQA methods have made substantial progress, challenges persist in addressing the diversity of distortions and ensuring adaptability to evolving image processing techniques. Techniques that effectively combine both global and local information, integrating attention mechanisms and feature fusion, tend to perform well across different datasets. Future advancements in NR-IQA may focus on refining attention mechanisms, exploring innovative feature fusion strategies, and adapting to emerging distortions.

3) COMPARISON RESULTS FOR NR-IQA TECHNIQUES ON AUTHENTIC DATASETS

Table 5 presents a comprehensive performance comparison of various NR-IQA methods on authentic datasets, including CLIVE, CID2013, KonIQ, RBID, SPAQ, and PaQ-2-PiQ. Several trends and patterns can be observed from the overall performance. Methods such as DVT-IQA, TempQT, and ASCAM-Former exhibit reasonably consistent performance across multiple datasets. For example, DVT-IQA achieves SROCC scores of 0.862 on CLIVE and 0.882 on CID2013, while TempQT shows SROCC scores of 0.870 on CLIVE and 0.903 on KonIQ. This consistency may indicate a robustness in handling various types of distortions present in authentic images. The ability to maintain performance across diverse datasets is crucial for real-world applications where images can vary significantly. While some methods show strong performance on certain datasets, they might face challenges in generalizing across diverse datasets. For instance, HiFFTiq and MUSIQ exhibit strong performance on KonIQ but comparatively lower performance on other datasets. HiFFTiq achieves a high SROCC of 0.922 on KonIQ but lower scores on other datasets, while MUSIQ performs well on SPAQ with an SROCC of 0.917 but shows variability on other datasets.

TempQT appears to perform well on the SPAQ dataset, which includes images captured using different smartphones. This suggests that TempQT might excel in handling temporal distortions commonly found in videos or sequences of images. The method could incorporate techniques to effectively assess quality in scenarios where temporal factors contribute to perceptual differences. DEIQT shows competitive performance, especially on the RBID dataset that focuses on real blur distortions encountered in everyday scenarios. The method might incorporate specific strategies to address challenges posed by real-world blur, distinguishing between motion blur, complex motion blur, and out-of-focus blur. MSTRIQ and HiFFTiq demonstrate strong performance, suggesting that a hybrid approach combining different techniques may contribute to their success.

PaQ-2-PiQ, being an extensive and fine-grained dataset with quality labels for both entire images and individual patches, presents an opportunity for assessing models' performance at different levels of granularity. Results shows that this dataset is a more challenging dataset. The overall trends indicate that certain methods excel in specific scenarios, while others demonstrate versatility across various datasets. Models like ASCAM-Former and ARET-IQA show competitive performance, with ASCAM-Former achieving an SROCC of 0.630 and a PLCC of 0.722 on PaQ-2-PiQ. The choice of the most suitable NR-IQA method may depend on the specific characteristics of the dataset and the nature of distortions present.

VII. DISCUSSION

A. IMPACT OF TRANSFORMERS IN IQA

Transformers have made a great impact on the field of imagine representation because of their complex architecture and attention mechanisms. Transformers provide a fine-grained representation of images by capturing complex spatial connections through the hierarchical levels of self-attention. This depth facilitates the modeling of complex relationships and extraction of high-level information critical for evaluating perceptual quality.

Positional encodings can help compensate for transformers' lack of intrinsic positional information. Methods such as sinusoidal positional encodings improve the transformer's comprehension of the pixel layout [118]. This is especially important for IQA, as human perception of image quality is heavily influenced by spatial layouts. Transformers can attend to multiple locations at once via multi-head attention technologies [119]. This allows the model to capture perceptual hierarchies in the context of IQA, ranging from small details to global structures. A rich collection of hierarchical representations that are consistent with human visual processes are produced by the specialization of each attention head in identifying particular features.

With the incorporation of Transformer models into IQA, a new age marked by unparalleled representational capability has begun. Transformers are superior to conventional techniques in capturing complex patterns and long-range relationships in images, which are crucial for evaluating perceptual quality. Empirical evidence from models like UniDASTN and MANIQA shows their remarkable performance across diverse datasets, effectively handling a wide array of distortions. Their adaptability to both local and global distortions positions them as versatile solutions for real-world IQA applications.

The inherent self-attention mechanisms of transformers play a pivotal role in their success. These mechanisms allow models to focus on relevant image regions, enabling them to understand and prioritize perceptually significant information. The attention to both local and global features

TABLE 5. Performance comparison of NR-IQA methods on authentic datasets. (The bold values indicate the highest performance, while the underlined values represent the second-highest performance.)

Method	CLIVE		CID2013		KonIQ		RBID		SPAQ		PaQ-2-PiQ	
	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC
DVT-IQA	0.862	0.882	-	-	-	-	-	-	-	-	-	-
TempQT	0.870	0.886	-	-	0.903	0.920	-	-	-	-	-	-
TFIQA-NR	0.863	0.894	-	-	0.921	0.933	-	-	-	-	-	-
CCL-IQA	0.8542	-	-	-	0.8905	-	-	-	-	-	-	-
TReS	0.846	0.877	-	-	0.915	0.928	-	-	-	-	0.554	0.625
HiFFTiq	-	-	-	-	0.922	0.936	-	-	0.920	0.925	0.620	0.712
LTMHPN-IQA	-	-	-	-	0.818	0.831	-	-	-	-	-	-
TRIQ	-	-	-	-	0.882	0.893	-	-	0.857	0.848	-	-
TranSLA	0.873	0.900	-	-	0.915	0.931	-	-	-	-	-	-
DEIQT	0.875	0.894	-	-	0.921	0.934	-	-	0.919	0.923	0.571	0.663
LoDa	0.876	0.899	-	-	0.932	0.944	-	-	0.925	0.928	0.578	0.679
ASCAM-Former	0.887	0.894	-	-	0.922	0.935	-	-	-	-	0.630	0.722
MUSIQ	0.702	0.746	-	-	0.916	0.928	-	-	0.917	0.921	**0.646**	**0.739**
ARET-IQA	0.882	0.899	-	-	0.932	0.945	-	-	0.924	0.932	-	-
CVC-IQA	0.872	0.891	0.885	0.896	0.915	0.941	-	-	**0.925**	0.929	-	-
MSTRIQ	-	-	-	-	**0.946**	**0.954**	-	-	-	-	-	-
VT-IQA	0.861	0.886	**0.944**	**0.947**	0.924	0.938	0.839	0.878	0.924	**0.938**	-	-
METER	**0.893**	**0.923**	-	-	0.929	0.937	**0.899**	**0.912**	-	-	-	-

contributes to a more nuanced evaluation of image quality. Transformer-based models, such as UniDASTN and VTAMIQ, showcase remarkable performance across diverse datasets encompassing a wide array of distortions. Their adaptability to both local and global distortions positions them as versatile solutions for real-world IQA applications.

B. CHALLENGES IN TRANSFORMER-BASED IQA

1) DATASET BIAS AND FINE-TUNING STRATEGIES

The sensitivity of transformer-based models to dataset characteristics poses challenges in generalization. Techniques such as domain-specific fine-tuning and adversarial training can mitigate dataset bias [120]. Adapting pre-trained models to recognize specific distortion characteristics enhances their ability to discern subtle variations in perceptual quality.

2) COMPUTATIONAL COMPLEXITY

Transformer computational complexity necessitates creative solutions, especially the quadratic attention complexity. Emerging methods such as kernelized self-attention [121] and sparse attention mechanisms [122] help maximize computational efficiency while preserving the model's ability to capture long-range dependencies. In resource-constrained contexts, transformer-based IQA must effectively balance representational power and computational efficiency.

3) GENERALIZATION TO UNSEEN DISTORTIONS

Despite their abilities, transformers may struggle with generalizing to unforeseen distortions. The dynamic nature of image processing technologies introduces new challenges, and ensuring the adaptability of transformers to emerging distortions remains an ongoing concern. The difficulty of applying transformers to unexpected distortions calls for ongoing learning techniques. Methods like transfer learning across different datasets and meta-learning, which helps models quickly adjust to new tasks, can make the model resilient in situations that are dynamically changing.

C. POTENTIAL SOLUTIONS

1) ATTENTION REFINEMENT STRATEGIES

Refining attention mechanisms is key to addressing challenges in localizing perceptually relevant information. Dynamic attention mechanisms that adapt to different distortions, or even hybrid attention mechanisms that combine spatial and channel-wise attention, can enhance the model's perceptual sensitivity.

2) DATASET AUGMENTATION AND DIVERSITY

Resilient dataset augmentation techniques can be used to address dataset sensitivity. By adding a variety of distortions to datasets, models are able to learn a wider range of features, which improves their ability to generalize.

3) FUSION OF MULTI-MODAL DATA

The fusion of multi-modal data, such as the combination of image and video information, presents exciting possibilities. Transformers, with their ability to model temporal dependencies, can benefit from joint training on diverse data modalities. This approach opens avenues for comprehensive

perceptual IQA, particularly in scenarios where temporal aspects notably impact image quality.

4) GENERATIVE ADVERSARIAL NETWORKS (GANS) AND ADVERSARIAL TRAINING

As GANs continue to introduce novel distortions, adversarial training strategies become crucial. Training transformers in an adversarial manner against GAN-generated distortions enhances their robustness. Adversarial training can be integrated into the training pipeline to ensure models adapt to the evolving landscape of image processing technologies.

D. FUTURE DIRECTIONS

1) EXPLORING CAPSULE NETWORKS AND TRANSFORMER HYBRIDS

Capsule networks focus on part-whole relationships, offering a complementary approach to transformers. Investigating hybrid architectures that integrate capsule networks with transformers could leverage the strengths of both paradigms, providing a more comprehensive framework for perceptual IQA. For example, employing capsule networks to enhance local feature extraction while utilizing transformers for global context may improve the accuracy of quality assessments.

2) META-LEARNING FOR ADAPTIVE IQA

Meta-learning holds promise for facilitating rapid adaptation to new tasks in IQA. Developing meta-learning frameworks that enable transformers to learn from diverse distortions and quickly adapt to new datasets can contribute to the continual improvement of model performance. This approach would enable transformers to maintain high accuracy even in the face of evolving image processing technologies.

3) QUANTUM-INSPIRED COMPUTING FOR IQA SCALE-UP

The exponential growth of data necessitates scalable solutions. Quantum-inspired computing architectures represent a frontier for scaling transformer-based IQA. Investigating the potential of quantum-inspired algorithms for transformer inference and training could enable the handling of larger datasets and more complex models, ultimately improving the efficiency and effectiveness of IQA systems.

VIII. CONCLUSION

In conclusion, the integration of Transformer-based models in IQA marks a paradigm shift, unleashing unprecedented capabilities in capturing complex spatial dependencies and perceptual hierarchies. While the research has been accompanied by challenges such as dataset biases and computational complexities, the field has responded with innovative solutions, refining attention mechanisms and adapting to emerging distortions. Looking forward, the future of IQA lies in a convergence of AI paradigms. Hybrid architectures, multi-modal data fusion, and adversarial training are poised to elevate transformers beyond mere assessors, transforming them into intelligent networks capable of adapting to the dynamic landscape of image processing technologies. Together with transformers, capsule networks, meta-learning, and quantum-inspired computing promise not just new technological developments but also a new era of intelligent perception. The potential is vast, and the transformer, with its intricate architecture, heralds the dawn of an era where machines perceive, learn, and transcend the boundaries of conventional assessment.

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